

A Parametric Design Approach for Affordable Customized 3D Socket for Transradial Upper Limb Prostheses

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Abstract—In upper limb prosthetics, customization of prosthetic designs is necessary to adapt them to each user's unique anatomical dimensions and ensure comfort and functionality. In addition, streamlining the process is essential to reduce the waiting time for people in need. In this context, an approach to parametric and transradial prosthesis design, which emerges as the primary tool to meet these challenges, is presented. In this way, the step-by-step process of how this paradigm is used in conjunction with other design techniques is detailed, thus obtaining a prosthetic stump-forearm interface and a forearm socket fabricated by 3D printing with PLA material. Likewise, customizing these elements guarantees a natural and realistic imitation of a forearm, which leads to the user feeling a sense of belonging toward the prosthesis itself. The results of this study show the fabrication of the final prosthetic design and its respective preliminary evaluation on a volunteer user with a long transradial congenital amputation condition based on the anthropometric measurements described in the design process. In addition, this process can optimize the design time compared to other prosthetic procedures, facilitating their productivity by manufacturing. This process, optimized in design time and computational load, benefits the creation of affordable prosthetic devices and sets a precedent in parametric design research, with potential for future applications in biomedical innovations.

Index Terms—CAD design, parametric, prosthesis, socket, transradial, upper limb.

I. INTRODUCTION

Technological advancements have led to the development of upper limb prostheses incorporating advanced technologies, introducing complexity that directly impacts the manufacturing time [1], [2]. In upper limb prostheses, tools, and design techniques to simplify this process have emerged, with parametric design being a noteworthy solution [3]. By embracing this paradigm, the project duration is reduced, and greater customization is introduced. This approach facilitates a more precise adaptation of devices to the user's residual limb or



Fig. 1. Prosthetic socket performing the arm flexion movement.

amputated area, fostering a more comfortable and natural experience, thereby promoting greater user acceptance [4], [5].

In this context, transradial myoelectric prostheses are technologically advanced devices placed on the arm of a person with an upper limb amputation condition in the wrist or forearm region, designed to be controlled by electrical signals generated by the user's muscles. These devices are characterized by their ability to detect electrical signals generated by the user's residual muscles, allowing for more inherent control of the device [6]. Although integrating advanced technologies adds complexity to the design of these prostheses, directly influencing the required manufacturing time [1], it is estimated that, based on similarities with previous projects [7], [8], the process could take approximately 20 hours. This finding underscores the urgent need to significantly streamline and optimize this process using parametric design [7], [9].

Research concerning the manufacture of prostheses emphasizes that parametric design promotes consistency and replicability [7], [8]. Establishing specific parameters ensures coherence in the final development, essential for large-scale manufacturing and component standardization, positively impacting production efficiency and costs [7]- [11]. Furthermore, precision in adjusting different prosthesis parts in response to user feedback is highlighted [12], [13], [17]. Parametric modifications are executed more efficiently, allowing for faster and more specific expert adaptation. Additionally, parameterization simplifies incorporating technological advancements

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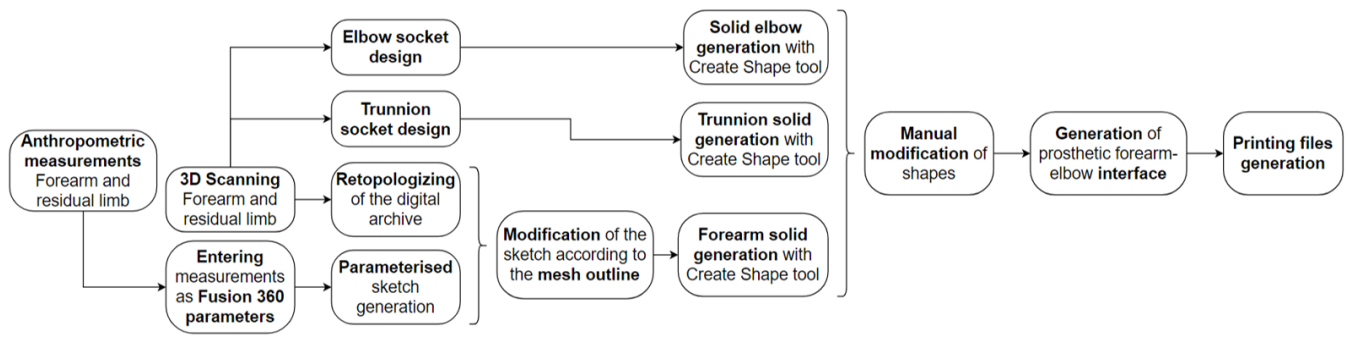


Fig. 2. Flowchart for the prosthesis design process.

and continuous improvements, as changes are implemented more promptly [11]- [14].

This study aligns with technological advancements by exploring the potential and benefits of parametric design in developing upper limb prostheses, specifically myoelectric transradial prostheses. The significance of customization in adapting to the individual needs of each user is emphasized, and this technique optimizes the design process while maintaining a scientific and rigorous approach. Despite parameterization, the need for fine manual adjustments is recognized to optimize both the prosthesis and the prosthetic socket, aligning them perfectly with individual needs. Nevertheless, this work presents the step-by-step application of parametric design, from taking measurements to obtaining the final prosthetic model in Fusion 360 software. The result, shown in Fig. 1, is a personalized and adjustable prosthetic socket 3D printed for evaluation, ensuring the user's precise fit and functionality. Thus, the aim is to contribute significantly to optimizing the design of upper limb prosthetic devices.

II. METHODS

This section details the design process of the transradial myoelectric prosthesis using parametric design techniques in Fusion 360 software. Thus, the flowchart presented in Fig. 2 guides the steps from the initial phase of the process, which comprises the scanning of the individual's upper limb, to the culmination of the process, with the final fabrication of the prosthetic socket. Likewise, said socket consists of the design of the prosthetic forearm, where anthropometric measurements are used to parameterize the solid; the section for the residual limb, where a Freeform with this shape is added and joined with the forearm; and the forearm socket, where the petals are formed and used to attach a fitting mechanism. It is

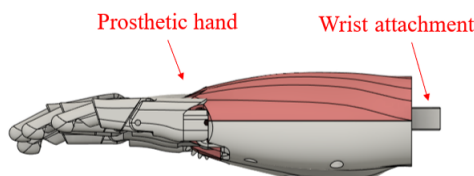


Fig. 3. LIBRA NeuroLimb 3D wrist and prosthetic hand design.

relevant to note that the LIBRA NeuroLimb 3D hand and wrist design, shown in Fig. 3, are used as a reference to develop the prosthetic forearm [15], in this case, for a 32-year-old Peruvian male with a congenital long right-sided transradial congenital condition.

A. Prosthetic forearm design

The following describes the parameterization process for designing the prosthetic forearm, aiming to optimize the design time while considering factors such as lightweight construction, mechanical properties, and customization. With parameterization, there's no need to start a design from scratch; instead, references can be utilized at each step. This approach allows more time dedicated to personalized modifications since it begins with a predefined shape based on the user's anthropometry. First, the values in Table I are entered as user parameters in Autodesk Fusion 360. These values correspond to the anthropometric measurements taken from a user with a transradial congenital amputation. Fig. 4 illustrates how these values are applied to parameterize the forearm sketch. Note that 72 mm is subtracted from the arm length to accommodate the petal stroke from the forearm socket, avoiding collisions when flexing the forearm, as will be explained in the 'Forearm socket design' subsection. The above measurement is defined as the length of the smallest petal, so it should be the same for each design unless the length is changed.

TABLE I
USER PARAMETERS.

Name	Unit	Expression	Value
D1	mm	34.6 mm + Offset	35.10
D2	mm	32.5 mm + Offset	33.00
D3	mm	30 mm + Offset	30.50
D4	mm	29.42 mm + Offset	29.92
Forearm_length	mm	151 mm - Petal_length	79.00
D_length	mm	Forearm_length / 3	26.33
Offset	mm	0.5	0.50
Thickness	mm	3 mm	3.00
Petal_length	mm	72 mm	72.00

Next, a second sketch is created, shown in Fig. 5. This defines the sweep trajectory that will generate the solid from the previous sketch. This new sketch is also parametric and

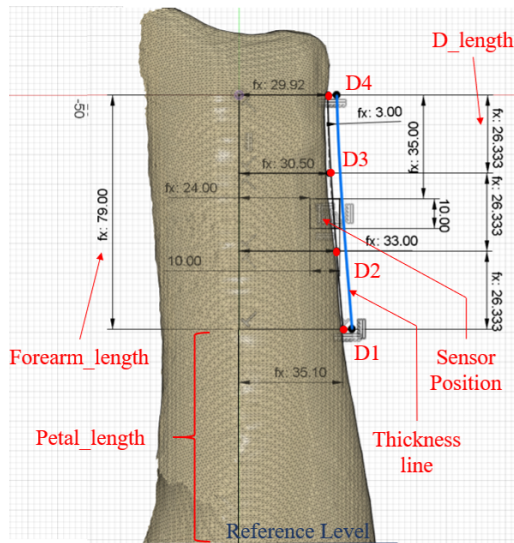


Fig. 4. Parametric forearm sketch.

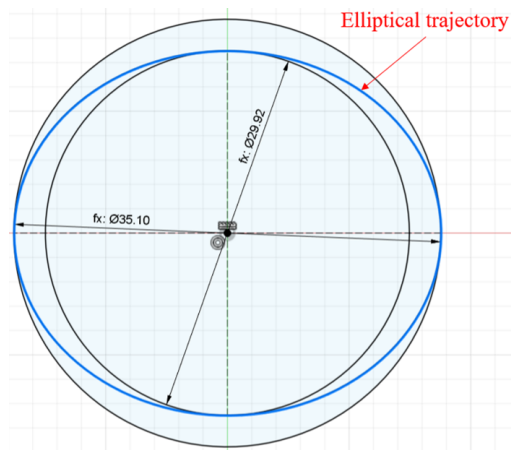


Fig. 5. Parametric elliptical trajectory.

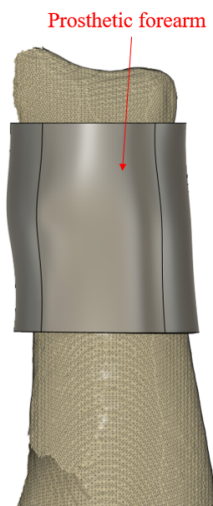


Fig. 6. Modified parametric prosthetic forearm.

configured to make the trajectory elliptical, considering the parameters D1 and D4. The purpose is to generate a solid similar to a natural forearm with the described trajectory. Then, using the "Create Shape" tool (Freeform), adjustments and modifications are made to perfect the solid obtained, guaranteeing an accurate design adapted to the user's specific needs. Thus, the shape shown in Fig. 6 is obtained, allowing the design of the residual limb to be carried out.

B. Residual limb design

Fig. 7(a) shows a sphere-shaped Freeform used to create the residual limb. This is cut in half, and then the "Pull" tool is applied to obtain the desired shape of the die. This tool adapts the sphere's geometry to the nearest surface, so the only near surface should be the forearm scan. This element is then adjusted to establish an offset of 0.5 mm between the die and its socket, facilitating socket removal during the first placement and contributing to user comfort by allowing skin ventilation.

Weld vertices must be generated to join the prosthetic forearm to the residual limb. Both elements must have the same number of vertices for a successful connection. Then, in this case, the surface must be thickened to generate a solid of 3 mm. This thickness is selected to provide greater robustness in the mechanical properties and to obtain a weight that does not exceed the one mentioned in [17], referring to the average weight of transradial prostheses. In addition, PLA plastic is used with the FDM fabrication method since the properties of such material allow for obtaining a prototypical version of the 3D design. This can be shown in Fig. 7(b).

Fig. 8 presents the final details, referencing the LIBRA NeuroLimb prosthetic hand to design the socket that connects this hand to the forearm-wrist interface. A nerve is incorporated along this interface to improve its malleability after fabrication with PLA plastic, providing greater comfort to the user. This is designed to create a space that, due to the properties of the plastic, can deform to increase pressure (for a tighter fit if

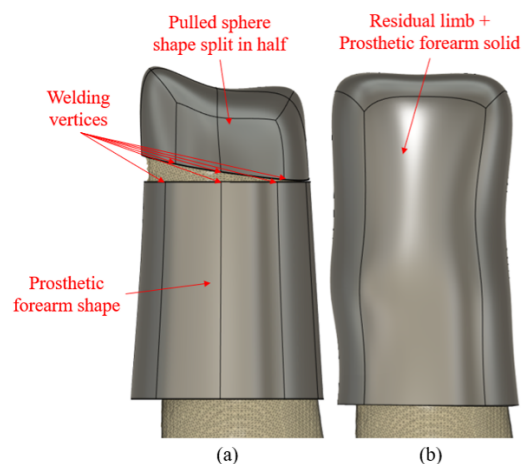


Fig. 7. Design process of the entire prosthetic forearm: (a) Punched sphere and prosthetic forearm shapes (b) residual limb and prosthetic forearm interface.

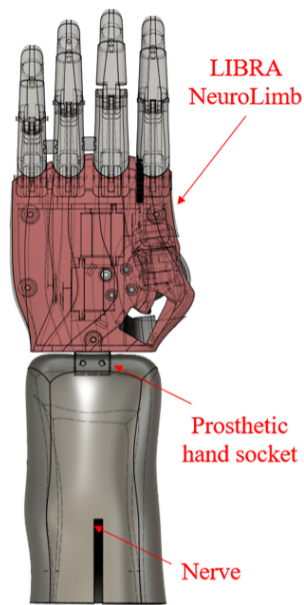


Fig. 8. Final details of the forearm-residual limb socket interface concerning LIBRA NeuroLimb hand.

needed) or remain as is (if the arm has been inserted without issues). It also allows minimal effort when inserting the arm, as the nerve can be slightly separated to facilitate the process.

C. Forearm socket design

The forearm socket is designed using Freeform shapes to achieve a geometry of petals that extend to form a disc, similar to the end of the prosthetic forearm. This geometry is designed to fit the shape of the socket. A threaded adjustable mechanism, shown in Fig. 9(a), is incorporated into the petals to attach to the user's limb, allowing flexibility due to the PLA composition. The length of the shortest petal is limited to 72 mm, taken as the "petal_length" reference for the prosthetic forearm design to avoid movement restrictions and ensure user comfort. The other petals have lengths that allow a uniform adaptation of the adjustment mechanism, avoiding tension in the thread. The longest petal, which extends below the elbow, facilitates full arm flexion without affecting the adjustment mechanism.

The forearm socket design is conceived as a 20 cm long template, although only 13 cm is required from the longest petal. This is done to provide a wider working area and ensure continuity when cutting the piece, facilitating the assembly of the components. As shown in Fig. 9(b), the starting face of the socket is guaranteed to be uniform with the final face of the forearm, ensuring aesthetic and functional continuity. This means it is unnecessary to redesign the socket from scratch for each user, as the modifications are made from the template design.

III. RESULTS

This section presents the results obtained through parametric and Freeform design techniques. The process described in

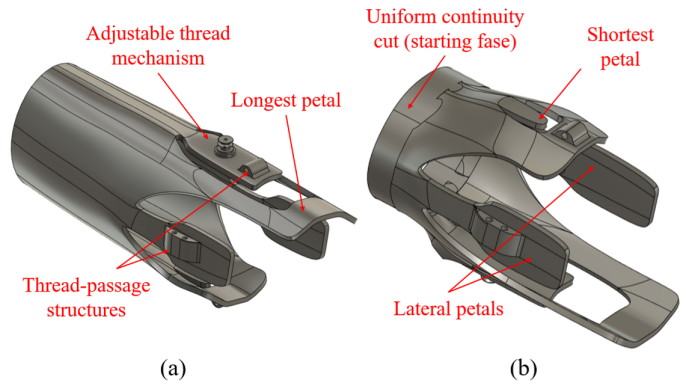


Fig. 9. Forearm socket: (a) Generic model (b) Model modified according to user's anthropometry.

Fig. 2 is completed, from generating the complete prosthetic interface to its fabrication using the 3D printing additive manufacturing technique. It also includes the preliminary evaluation of the prototype, in this case, with the user for whom the anthropometric measurements were acquired. On the other hand, although a complete parameterization of the forearm socket was not achieved, a partial parameterization focused on continuity was implemented. This was achieved by parameterizing the geometry that extrudes the initial face to give it a continuous shape, which depends on the projection of the final geometry of the forearm.

Fig. 10 shows the overall design of the transradial prosthesis. Four holes have been incorporated into the prosthetic forearm and socket to insert wires of 1.4 mm in diameter and 2 cm in length. These wires are used as an alternative method of assembly due to the limitations in the bed size of the available 3D printers; therefore, the prosthetic socket and the forearm-elbow mechanism must be printed separately. However, they allow precise assembly of the components, eliminating any perceptible discontinuities and achieving a greater degree of realism in the appearance of the prosthesis.

The design time was reduced to 3 hours, considering short rest intervals. This demonstrates the efficiency of the described method compared to traditional methodologies, as described in [1] and [2]. Also, Table II shows the main printing parameters, considering that this has been done using the PrusaSlicer software for the Prusa i3 MK3 printer, with a printing duration for each part of approximately 7-8 hours. This time efficiency allows the reproduction of multiple designs within a standard 8-hour day and by need.

Printing only two pieces facilitates continuous manufacturing between days, promoting productivity as needed. Fig. 11 shows the prosthetic socket manufactured, which is already assembled and ready to be fitted.

Fig. 12 preliminarily evaluates the design by showing its correct fit on the user's forearm as it enters and exits the arm. If, after printing, the prosthesis does not fit perfectly, surface sanding processes can be applied to adjust it to the user's specific dimensions. This approach not only ensures a customized fit but also highlights the versatility of the design

TABLE II
CONFIGURATION OF BASIC PARAMETERS FOR PRINTING THE DESIGNED COMPONENTS.

Parameters/Components	Material	Layer Height (mm)	Wall line count	Infill density (%)	Support type	Support density (%)	Total time of printing	Total cost of printing
Residual limb - Forearm	PLA	0.2	2	60	Everywhere	5	7h 5m	10.49 (PEN)
Forearm socket	PLA	0.2	2	50	Everywhere	5	7h 44m	11.87 (PEN)

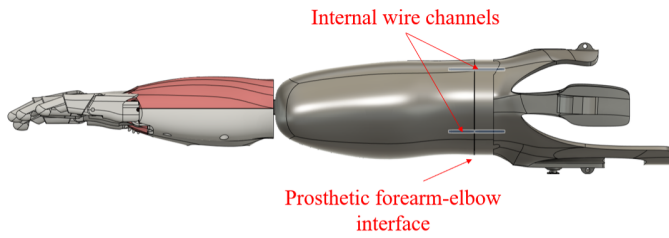


Fig. 10. Final assembly of the prosthetic interface (residual limb, prosthetic forearm, and forearm socket) placed in LIBRA NeuroLimb.

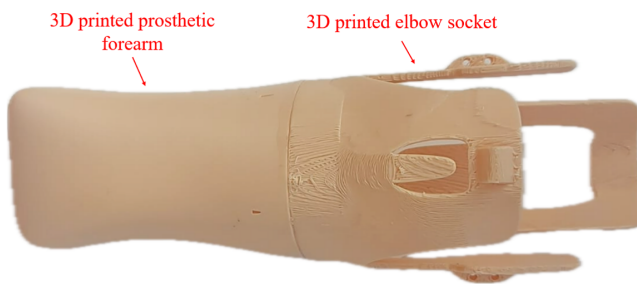


Fig. 11. 3D printed prosthetic socket assembly.



Fig. 12. Prosthetic socket attached to the arm.

to be adjusted according to individual needs, as this is achieved by the adjustment mechanism incorporated in the petals. Finally, the weight of the prosthetic socket (prosthetic forearm and forearm socket) is 103 g; even when incorporating the Libra NeuroLimb hand (including the electronic components), which weighs 188 g, the total would be 291 g. Despite being a prototypical version, such a result is significantly lighter than the average mentioned in [5] and [17], which is 640 g for transradial prostheses.

IV. DISCUSSION

In the current study, Autodesk Fusion 360 software has facilitated the design, parameterization, and subsequent file

generation required for 3D printing fabrication processes, particularly within upper limb prosthesis and component design. Its selection stems from a deliberate choice to leverage its inherent properties in this specialized context, wherein integrating design, parameterization, and fabrication workflows is paramount for achieving optimal outcomes.

Nevertheless, it is imperative to acknowledge certain limitations inherent to the software. Given its classification as a lightweight system in comparison to counterparts in computer-aided design (CAD), instances were observed where specific tools and functionalities, when applied to surfaces, either became non-operational or incurred disproportionately lengthy processing durations, issues that might have been swiftly mitigated within alternative software ecosystems. Additionally, the incapacity to parameterize measurements in Freeform format necessitated a recourse to parametric design via sketch modalities, with subsequent adaptation of organic geometries using Freeform commands. Challenges also arose during attempts to amalgamate designs produced in Freeform mode from disparate file sources. The absence of a dedicated tool for such integration necessitated an alternative approach involving projecting one piece's geometry onto another and establishing a connection interface based on this projection.

A review of extant literature reveals a prevailing reliance on elevation-based methodologies for parameterization endeavors. However, this conventional approach is characterized by inherent sluggishness and computationally intensive operations, often leading to discrepancies between successive planes. In contrast, the current study achieved more efficient parameterizations by leveraging scanned forearm geometries as references and employing Freeform commands to iteratively mold these geometries, thus ensuring bespoke fittings tailored to the unique characteristics of individual residual limbs.

The scope of the proposed parametric design framework for transradial amputee fittings necessitates a minimum amputation length of 7 cm measured from the elbow for applicability. This criterion ensures a suitably snug prosthetic fit and sufficient coverage for surface electromyography (sEMG) sensors, as well as ensuring user comfort through incorporating a robust security system and strategically positioned reinforcements. It should be noted that this technique is not limited to the design of transradial prostheses but can be applied in different approaches, covering both upper and lower limbs and even creating other types of devices that are not necessarily prosthetic.

Looking ahead, the aspiration is to extend the parametric design framework to encompass elbow fittings alongside comprehensive measurements spanning fingers, palm, wrist, and forearm dimensions. Realizing this objective entails the acquisition of supplementary arm and forearm scans, en-

compassing both amputated segments and intact contralateral limbs. Furthermore, exploring lightweight yet resilient materials such as Carbon Fiber via additive manufacturing techniques holds promise, albeit necessitating adjustments to fitting wall thicknesses commensurate with the mechanical properties of the selected material.

V. CONCLUSION

This research has demonstrated the feasibility and relevance of an innovative approach, combining CAD techniques, such as parametric design and Freeform shapes, to develop adjustable and customized prostheses for the upper extremities. Based on anthropometric measurements and previous 3D scans, the abovementioned techniques have generated an accurate and time optimized design methodology, achieving a prosthetic forearm and its respective socket, where both components fit perfectly to the user. Customizing them may provide a high sense of belonging to the person, as indicated in [1], [3], where the fit between the prosthesis and the natural forearm is fine-tuned. Traditionally, this has been challenging to achieve, manifesting the design process's intrinsic value.

This design approach opens the door to new opportunities in biomedical engineering by allowing the development of complex prosthetic devices in a shorter amount of time and ensuring accuracy of fit. Thanks to the parametric design obtained, customizing and adjusting 3D models, depending on the needs of each user, promises a significant advance in the implementation of a transradial prosthesis, in this case, tested on the LIBRA NeuroLimb prosthetic hand. Furthermore, this design approach can be used to innovate and optimize upper limb prosthetic devices and other parts of the body.

According to the results obtained and based on the documentation reviewed in [7], [8], the design time has been optimized, reducing it to only 3 hours. This efficiency allows working simultaneously with several users, speeding up the production of components. Reducing design time eliminates the need to use high-performance software for extended periods. This results in a more accessible process, as it does not require the prolonged use of demanding computational tools.

The design was preliminarily evaluated by fabricating the components using 3D printing and testing them on a volunteer subject. The results showed that the prosthesis's forearm and socket adequately fit their anthropometric measurements and are much lighter than in other results [5] and [17]. The choice of PLA plastic and the adjustment mechanism incorporated in the petals of the forearm socket allow the tightness to be regulated as needed. The optimal level of fit is achieved through this mechanism since this material can withstand light bending loads of 17.8 N and 10.1 N in the movements of flexion and extension, respectively, when its thickness is 3 mm [16].

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